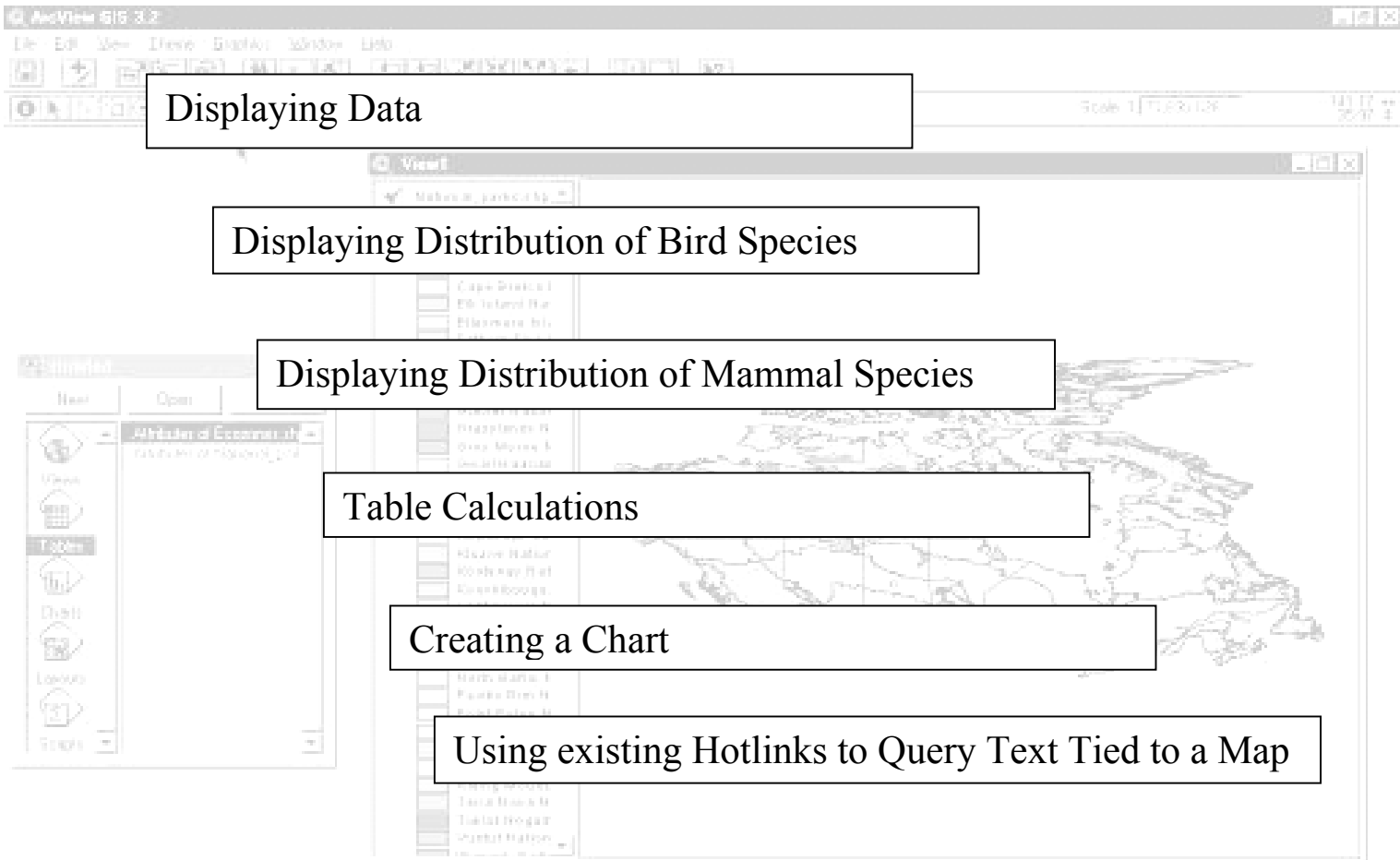


Wildlife and Species Richness of Canada



GIS Information

Here's a list and explanation of important features in ArcView:

Project : In ArcView, a file for organizing your work. Projects use five types of documents to organize information: views, tables, charts, layouts, and scripts.

View  : A component of an ArcView project used for displaying, querying, and analyzing geographic themes.

Table of Contents: All themes in a View are listed to the left of the map. The Table of Contents shows the symbols used to draw features in each theme. The check box next to each theme indicates whether or not it is turned on or off in the map (whether it is currently drawn on the map).

Theme: A set of related geographic features, such as streets, parcels, or rivers, and the characteristics (attributes) of those features.

Tables  : Information formatted in rows and columns. A component of an ArcView project used for displaying Tabular data.

Layout  : The design or arrangement of elements in a digital map display or printed map. A component of an ArcView project used for creating presentation-quality maps.

Hotlink: In ArcView, a way to display data (for example, a file, image, ArcView document, or project) directly from a view, by clicking on a feature.

ArcView Buttons Reference Guide

View - Buttons



View – Tools

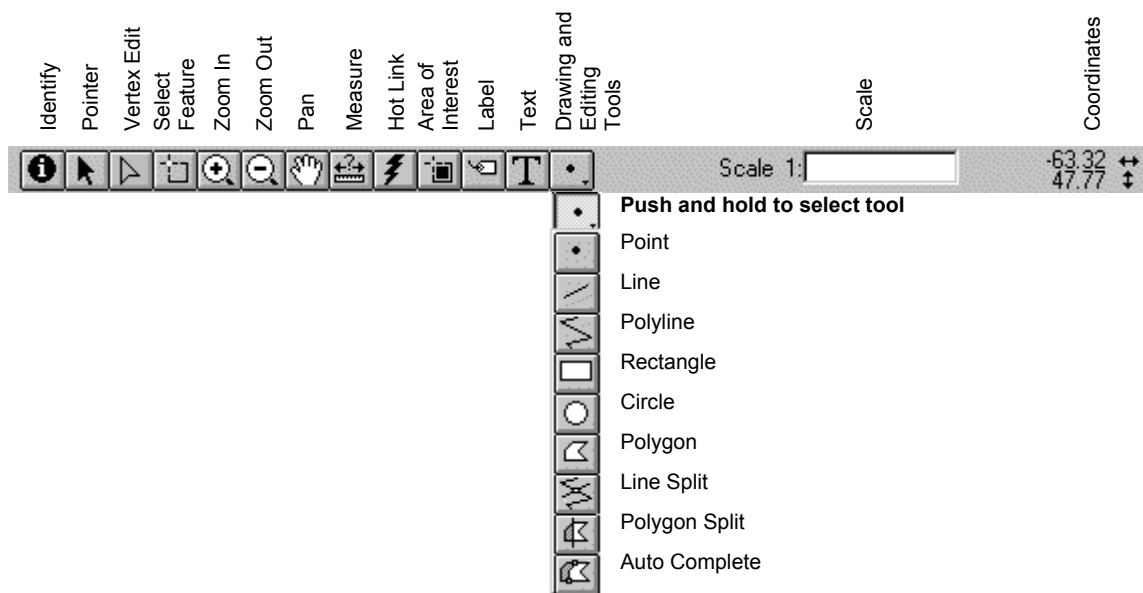


Table - Buttons



Table - Tools

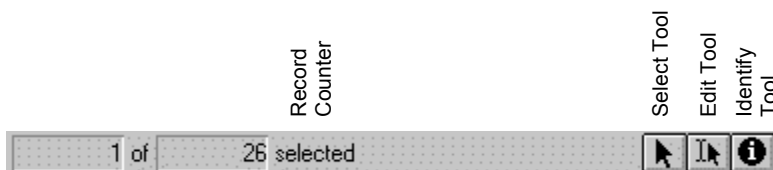


Chart - Buttons

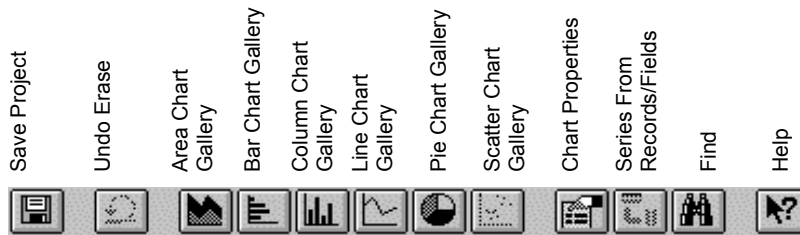
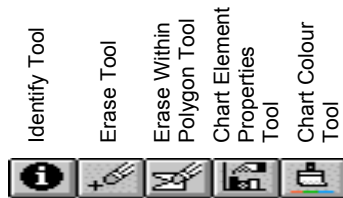
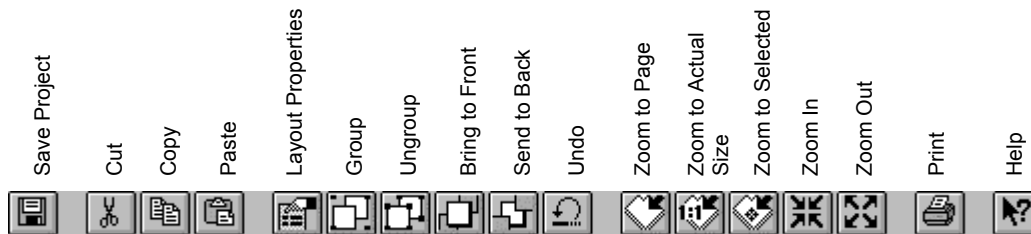


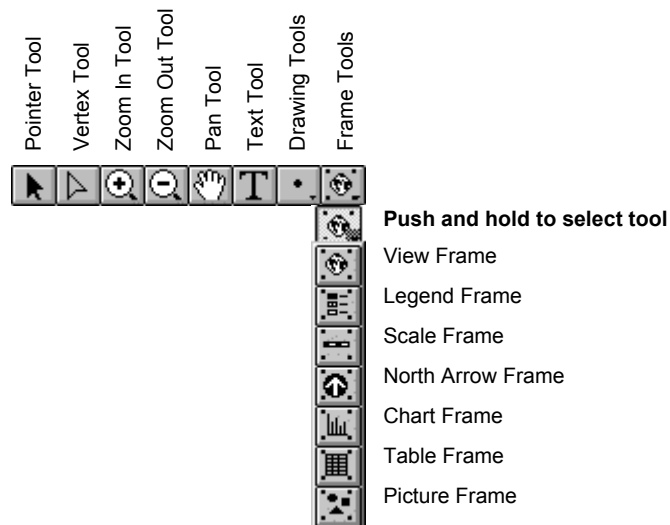
Chart - Tools



Layout - Buttons



Layout - Tools



For more Information: <http://www.esricanada.com>

Getting Started

Before you begin, the lesson data folder must be **copied** into a drive in which the student has writing and editing capabilities. This will probably be the drive containing your own personal account on the school's system or it may be the **C: drive** of the machine you are currently working on. If the latter is the case you will have to work on the same machine each day until this assignment is complete. Each school set-up is different so work with your teacher to prepare your data folder.

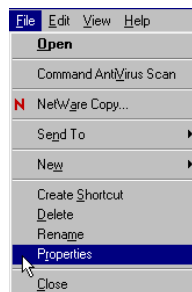
! (The data for the lesson must first be downloaded from this Web site and placed on the school computer or network. Ask your teacher where to find the data folder that has been downloaded).

To do this, use the file manager program on your computer. For Windows-based systems this will be **My Computer / Windows Explorer /**.

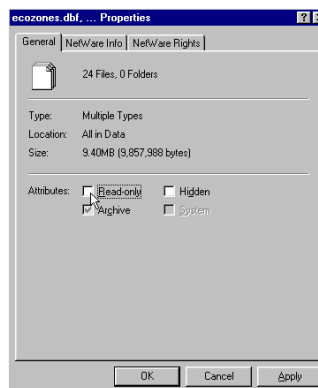
Once the data folder has been **copied** to the required destination the files must be made usable by removing the **read only** attribute. For Windows-based systems follow these instructions:

! (The data downloaded from the Web Site is set as **read only** so that it cannot be changed or altered by accident.)

1. Open your new data folder.
2. Choose **Edit > Select all** (they will all appear Blue)
3. From the menu choose:
File > Properties



4. Remove the check mark in the box beside the label **read only**.

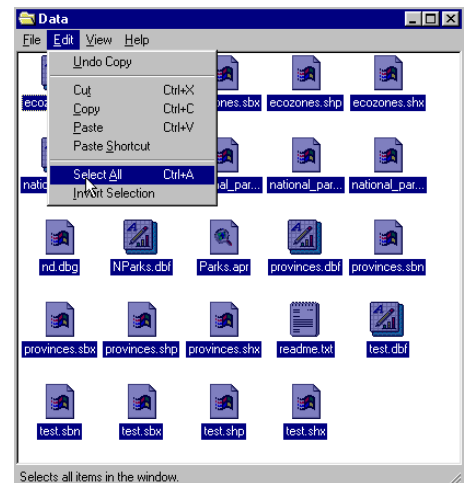


5. Click **Apply**, then **OK**.

The data in your new data folder can now be opened, edited and saved using ArcView or other GIS software that can use SHAPE files to carry out the following project.

! Denotes an observation.

? Denotes a question to be answered on the Student Activity Sheet.



STARTING THE PROJECT

This lesson will deal with displaying data, graphing, table calculations and hot links to text files.

STARTING ArcView

Begin by starting up the **ArcView** program. If you are not sure how to do this, ask your teacher.

1. Choose *as a blank project*.
2. Click **O.K.**

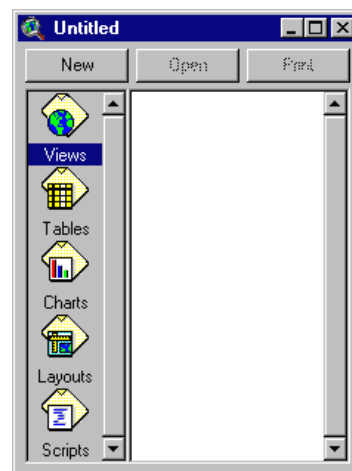


When the program starts, it automatically opens a project window titled "**Untitled.**"

If you like, you can maximize the **ArcView** window to fill the screen.

! In ArcView, you work within a project. A project is a collection of documents (map views, tables, charts, layouts, etc), document user interfaces, (called DocGUIs) and scripts. Only one project is open at a time.

3. Now you are ready to add documents to your new project.
4. With the "**Views**" icon selected, click on **New** and maximize the **View1** window.

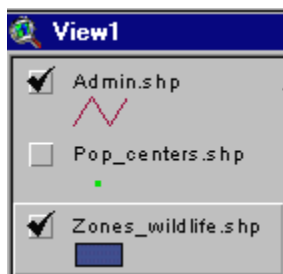


5. Click the **Add Theme**  button and navigate to the data folder where your data is stored. Hold down your **SHIFT** key and select the following files:

Admin.shp
Pop_centers.shp
Zones_wildlife.shp

6. Click **O.K.**

This will add these three themes to the table of contents of the new view. They are now available for display and editing.



7. Click and drag to move the themes up or down in the table of content to change the order of display if necessary.

! Note the themes are displayed from the bottom to the top so line and point themes should be on top of polygon files to be visible.

8. Click on the square next to the themes you want to display. The theme that appears raised in the table of contents is the current active theme. This can be change by clicking on other themes.

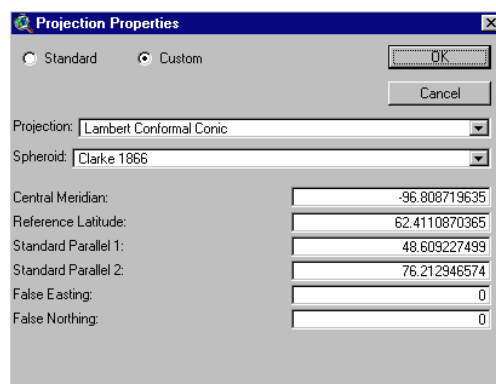
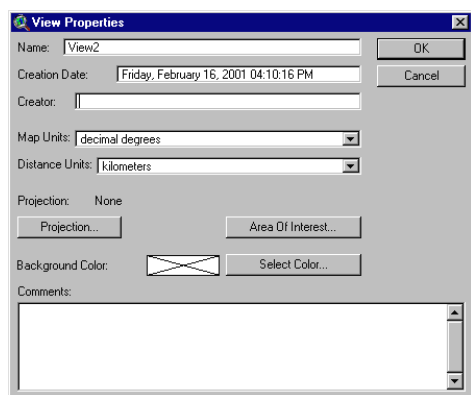
9. Remember to **Save** your project as you go along. Go to the **File** menu and select **Save Project**. Save your work to the appropriate directory (ask your teacher if you are not sure). Give your project a name that will identify your work. (The suffix **.apr** means that you are saving your work as a project that can be opened up in the future).

ArcView - Changing the Map Projection

! A **Map Projection** defines the relationship between the map coordinates and the geographic coordinates, (latitude and longitude). Our current display shows what a map of Canada looks like with only a geographic projection.

Now we will give our map “projection coordinates” to make it look more like the maps which are familiar to us.

1. From the **View** menu, choose **Properties**.
2. Select **Properties**.
3. Select **Map Units** list. **Change the map units to decimal degrees.**
4. Select **Distance Units** list. **Change the distance units to kilometers.**
5. Click the **Projection** button.
6. Select **Custom**.
7. Select **Lambert Conformal Conic** from the projection list.



! Take a look at the parameters for this projection. Parameters are the descriptions of how the projection was made. Note that a spheroid was used. (The earth is a slightly flattened sphere toward the poles).

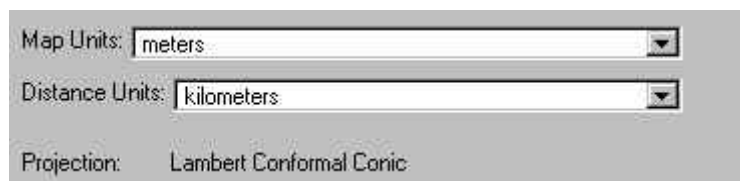
Conformal implies that all angles are indicated correctly. All angles measured on the earth's surface are given the same values on the map.

Conical means in the shape of a cone. In this projection, the relative positions on a globe are transferred to a plane in the shape of a cone around the globe. See how the longitudinal lines are closer together at the top of the image, that is, they are not parallel.

The **central meridian**, **false easting** and **false northing** are used to define the X (easting) and Y (northing) coordinates of the map. The **reference latitude** and **standard parallel 1 and 2** are used to determine the origin of the projection (i.e. the point from which you are viewing the globe).

1. Click **O.K.**

The bottom of the **View Properties** window should look like this:



9. Click **O.K.**

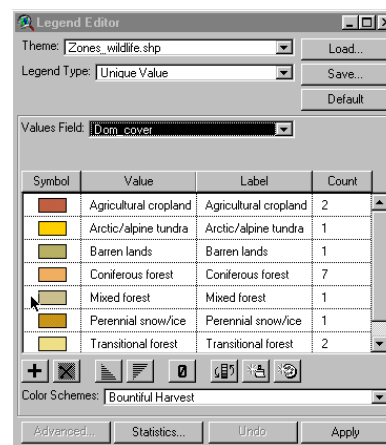
! Note the difference in the map presentation.

? How is it different from the non-referenced map?

ArcView - DISPLAYING DATA

Let's begin by displaying the dominant land cover and interpret its value as wildlife habitat.

1. Double click **zones_wildlife** theme.
2. Open **Legend Type** list.
3. Select **Unique Value**.
4. Open **Values Field** list.
5. Select **Dom_cover**.
6. Click **Apply**.
7. Close **Legend Editor** window.



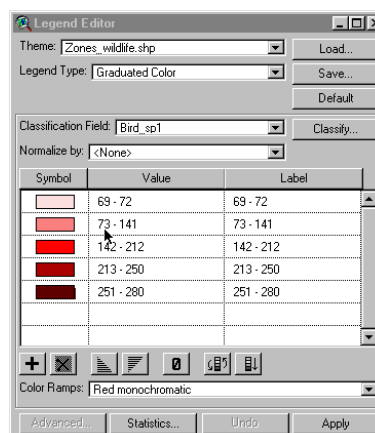
8. Click **Identify Tool**  and use the cursor to find an ecozone, which might have good habitat for each of the following:

- forest dependent birds
- small mammals that live in open agricultural fields
- caribou that calf on open tundra
- seabirds that nest on barren rocky cliffs

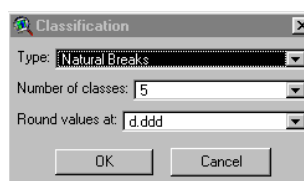
ArcView - DISPLAYING DISTRIBUTION OF BIRD SPECIES

Now let's make a map that shows the distribution of bird species by ecozone.

1. Double click **zones_wildlife** theme.
2. Open **Legend Type** list.
3. Select **Graduated Colour**.
4. Open **Classification Field** list.
5. Select **Bird_sp1** (number of bird species).



6. Click **Classify**.
7. Open **Type** list.
8. Select **Natural Breaks**. Click **O.K.**
9. Click **Apply**.
10. Close **Legend Editor** window.



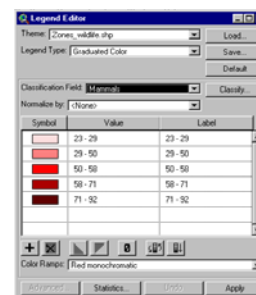
? What can you say about the distribution of bird species across Canada?

11. Using the **Identify Tool** , find the ecozone with the greatest number of bird species.

? Find the ecozone with the greatest number of bird species. Give some possible reasons why this ecozone has the greatest number of bird species?

ArcView - DISPLAY DISTRIBUTION OF MAMMAL SPECIES

1. Double click **zones_wildlife** theme.
2. Open **Classification Field** list.
3. Select **mammals**.
4. Click **Apply**.
5. Close **Legend Editor** window.
6. Using the **Identify Tool** , find the ecozone with the greatest number of mammal species.



? Find the ecozone with the greatest number of mammal species. Give some possible reasons why this ecozone has the greatest number of mammal species?

Repeat this process to determine the ecozone with the greatest number of trees (*tree_sp*) and the greatest number of reptiles & amphibians (*reptile_*).

? Determine the ecozone with the greatest number of tree species (*tree_sp*).

? Determine the ecozone with the greatest number of reptiles & amphibians (*reptile_*).

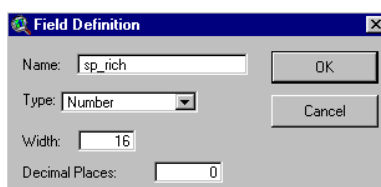
- ? Which ecozone has no tree species? Why?
- ? Which ecozones have no reptiles or amphibians? Why?

ArcView - TABLE CALCULATIONS

We are going to calculate the species richness of each ecozone. We will sum the reptile & amphibian, tree, rare plants, endemic plants, bird and mammal species values for each ecozone to get a total number of species in each ecozone.

1. Click **zones_wildlife** theme to activate it.

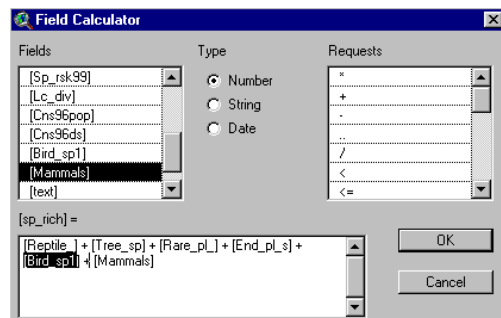
2. Click **Open Theme Table**  button.
3. From the MENU, click **Table**.
4. Click **Start Editing**.
5. From the MENU, click **Edit**.
6. Click **Add Field**.
7. Enter field name **sp_rich**.
8. Click **O.K.**



9. Click **Calculate**  button.

We will build an expression that will compute the species richness. To do this, double click on each component to be added to the expression. This will sum the selected columns. The expression should look like **[reptile] + [tree_sp] + [rare_pl] + [end_pl_s] + [bird_sp1] + [mammals]**.

10. Click **O.K.**

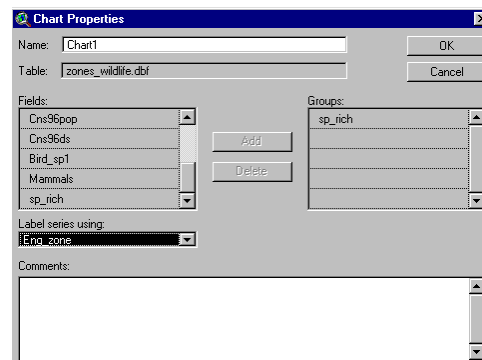


The calculated values should appear in the **sp_rich** field.

11. Click MENU item **Table**.
12. Click **Stop Editing**.
13. To save edits, click **Yes**.

ArcView - CREATING A CHART

1. Click **Create Chart**  button.
2. From **Field** list, select **sp_rich**.
3. Click **Add**.
4. Open **Label Series Using** list.
5. Select **eng_zone**.
6. Click **O.K.**



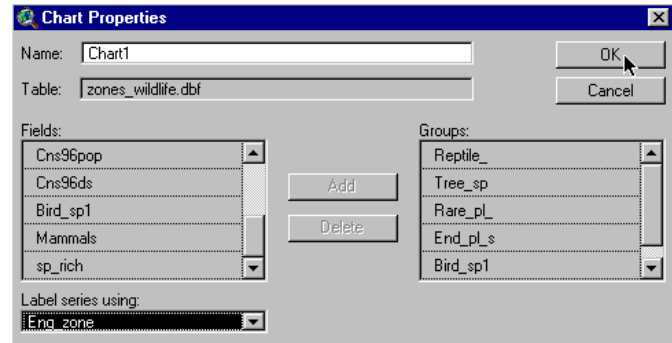
? From the graph you created in ArcView, which ecozone has the highest species richness?
Why?

7. Close the graph window.
8. From the **Table**, select **Mixedwood Plains** record.

Polygon 8 Mixed Wood Plains

9. Click **Create Graph**  button.

10. From **Fields** list, add **reptiles, trees, rare plants, endemic plants, birds** and **mammals** to groups.
11. Open **Label Series Using** list.
12. Select **eng_zone**.
13. Click **O.K.**

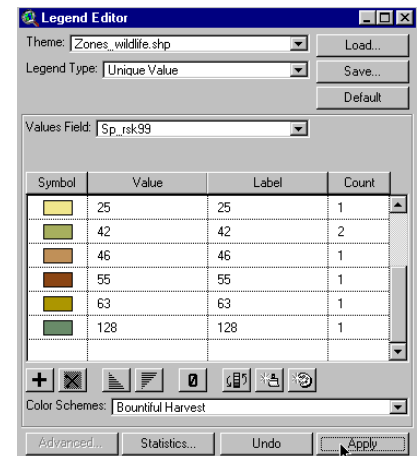


? Which species group makes up the largest percentage of the species richness in the **Mixedwood Plains** ecozone?

14. Close the graph window.
15. Activate the table and click the **Select None** button.
16. Close the table.

Now, we'll look at the species at risk in each ecozone.

1. Double click **zones_wildlife** theme.
2. Open **Legend Type**. Select **Unique Value**.
3. In the **Values Field** select **Sp_rsk99**.
4. Click **Apply**.
5. Close **Legend Editor** window.



? Which ecozone has the largest number of species at risk?
Why?

6. Check **pop_centers** theme to make it visible.

? Is there a correlation between species at risk and centers of high population?

ArcView - USING EXISTING HOT LINKS TO QUERY TEXT TIED TO A MAP

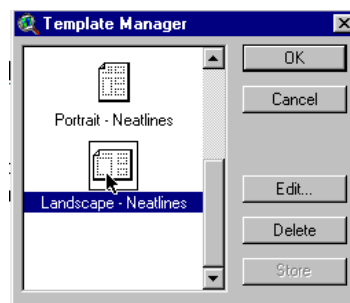
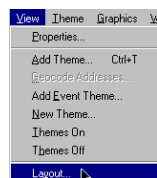
1. Click **Hot Link** button.
2. Move the  cursor to the ecozone you want to study. Click on it.
3. From the list of forest dependent species at risk that appears, select two.
4. Using the Internet, access the <http://www.speciesatrisk.gc.ca/Species/English/SearchRequest.cfm> Web site.
5. Research these two species and compile a fact sheet.

ArcView - MAP LAYOUT FOR PRINTING

Once you have a map classified and displayed to your satisfaction, you should prepare a map layout that can be used for printing. The purpose of this lesson is to set up your map on a page and annotate it for presentation.

SET UP MAP

1. Click Menu **View**.
2. Click **Layout**.
3. Select the **Layout Template** you want to use (One of the landscape templates is suggested) .
4. Click **O.K.**
5. Maximize the Layout screen.



EDIT TITLE

1. Double click on the title **View1**.
2. Edit the title to reflect the purpose of the map, e.g., forestry, and population.
3. Click **O.K.**
4. Resize the title using the corner handles and move it into a position that is visually balanced.
5. Click and drag to reposition the legend if necessary.

EDIT NORTH ARROW

1. Double click on the *north arrow* and select a different presentation. Ensure that the arrow is pointing in the representative north direction of the map.
2. Click and drag to reposition or rotate (using rotation function) as needed.

EDIT SCALE BAR

1. Double click the *scale bar*.
2. Choose a new style of scale bar. The scale bar has default properties so you will not be able to change everything to the way you want.
3. Change the units to **kms**.
4. Find a combination of style, scale and divisions that is visually pleasing.

5. Click and drag to position to an appropriate location.

ADD LABELS, TEXT, OTHER INFORMATION

1. Click on the **Text** tool.
2. Click at the position on your layout where you want the text to appear.
3. In the **Text Properties** box, enter your text, e.g., "produced by (your name, date)".
4. Select desired formatting information such as **alignment**, **spacing** and **rotation**.
5. Click **O.K.** when it appears, as you want it.
6. Click on **Pointer** tool.
7. Select the new text
8. Click once to change location or size. Click twice to change content.

! Experiment with all of the components of the layout until you have the look you want.

CHANGE THE NAME OF THE LAYOUT

You only have to do this if you want to save your project on the computer or on a disk.

1. Click menu **Layout**.
2. Select **Properties**.
3. Change the "name". Click **O.K.**

ARRANGING YOUR LAYOUT

! Experiment with the layout buttons (use the tools page provided at the beginning of this lesson if you are unsure of their function).

- to **group** (tie pieces together so they move as one)
- to **ungroup** (break apart the ties so that individual pieces move)
- **bring to front** (change order that pieces were laid down, e.g., lines on top of polygons)
- **send to back** (change order that pieces were laid down, e.g., polygons behind lines)
- **zoom** (make image bigger or smaller)
- **neat line** (border around layout)

PRINT MAP LAYOUT

1. Click **Print** button.
2. Click **Setup**.
3. Select your printer properties as outlined by your teacher.
4. Click **O.K.** to print.

! Note that the map can also be saved to a printer file.